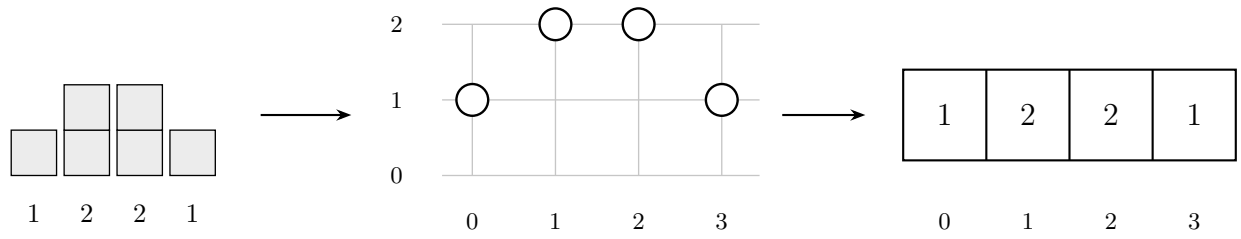
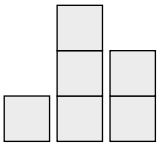
Heights are whole numbers, including 0. Neighbors differ by at most 1. Black circles and shaded boxes are fixed clues.

One marker in each column shows its tower height.

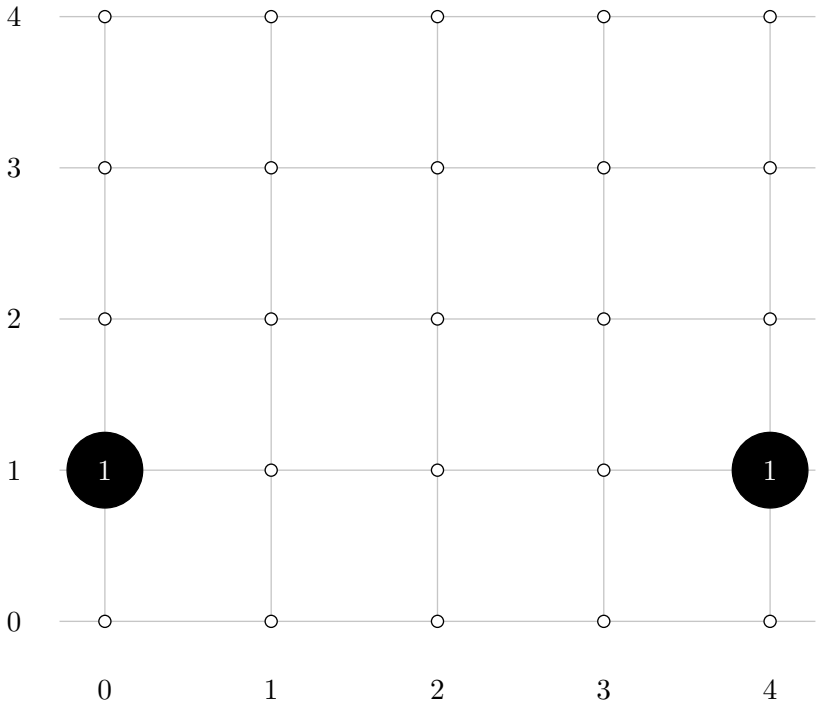


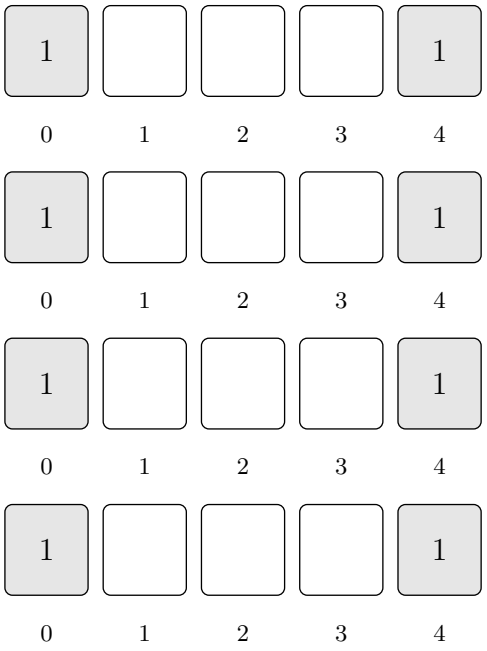
1 2 2 1
allowed



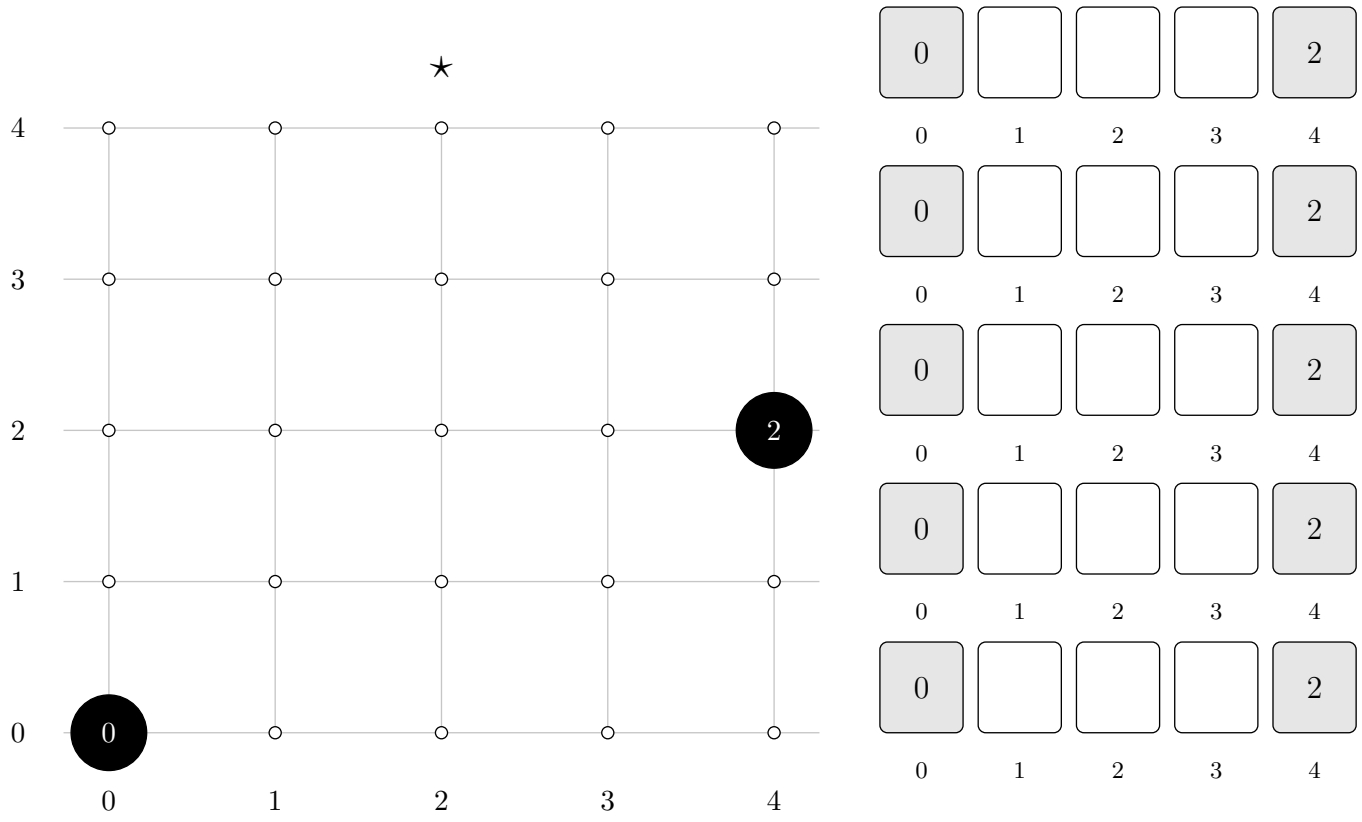
1 3 2
jump too big

Problem 1: Make four different landscapes with these clues. Can the middle height be 0? Can it be 3?

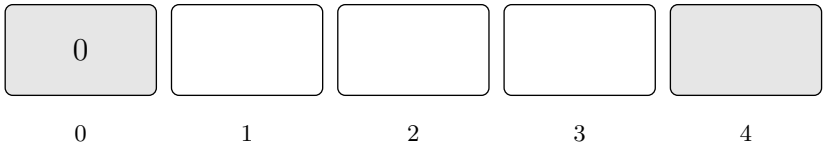




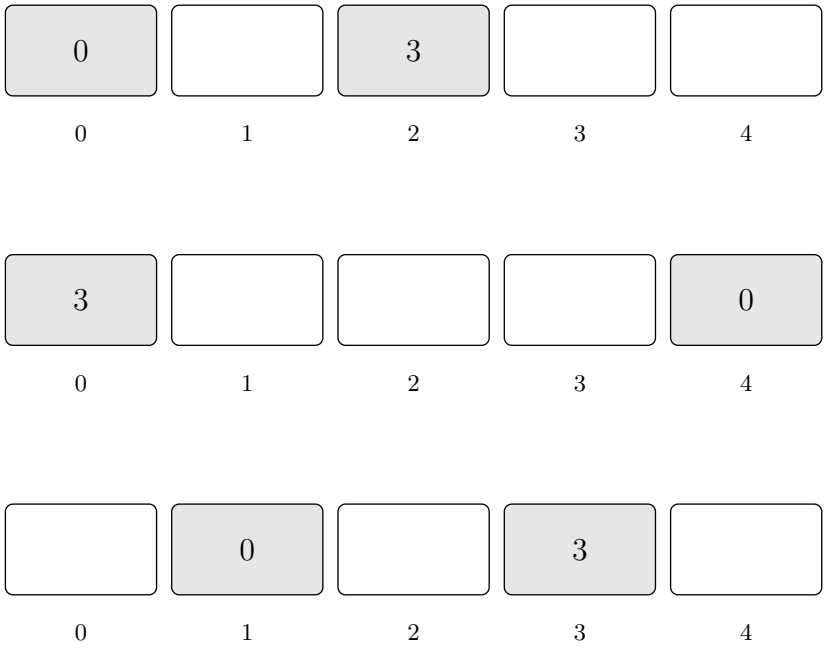
Problem 2: Which heights can the starred marker have? Make a landscape for each possible height and explain why the other heights cannot work.

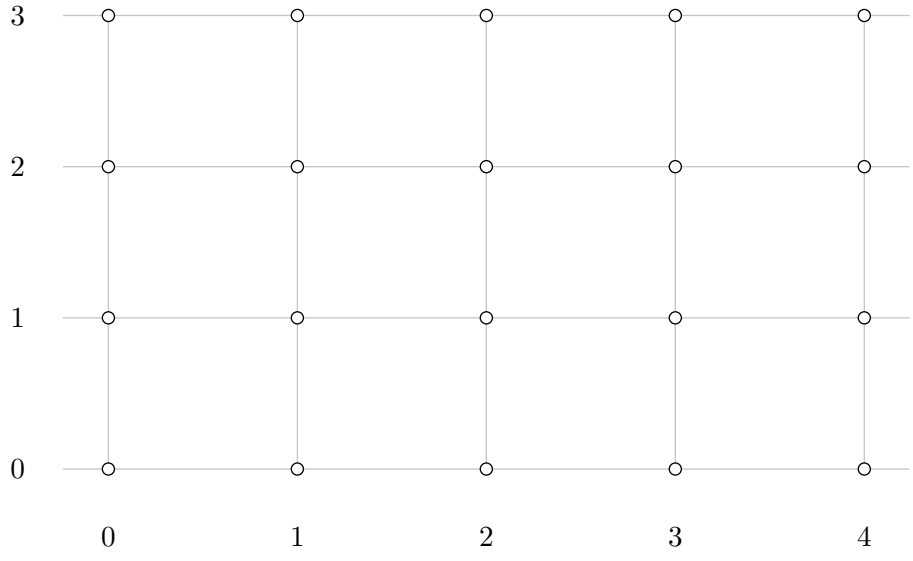


Problem 3: Keep the left clue at 0. Choose a height for the right clue that forces every marker. Find all such right-clue heights.

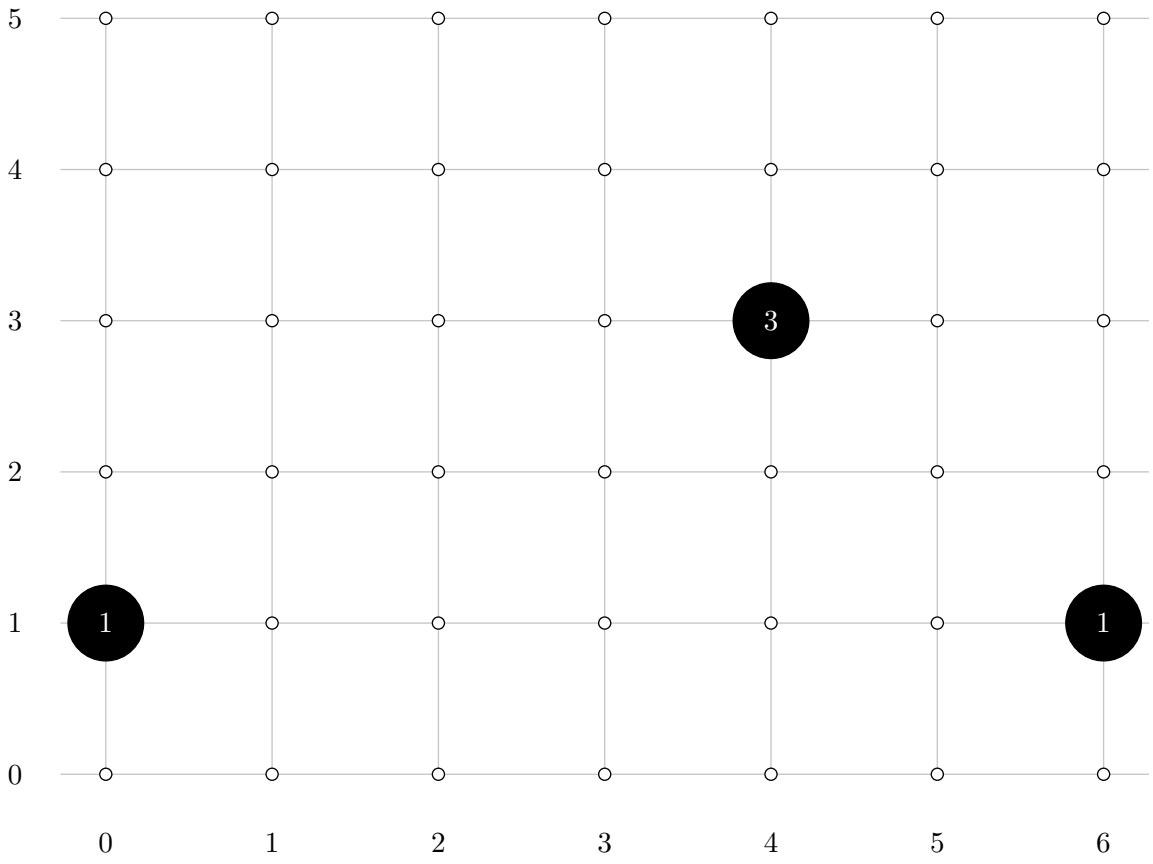


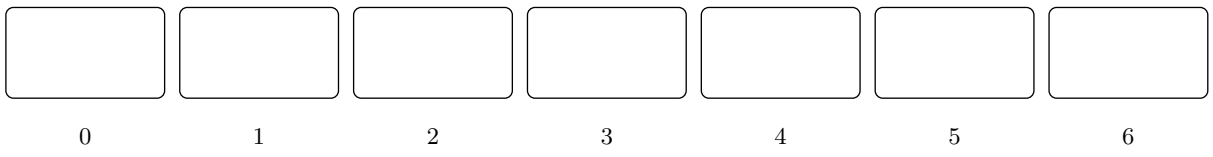
Problem 4: Decide which clue sets can be completed. Build the possible ones; explain what prevents the others.

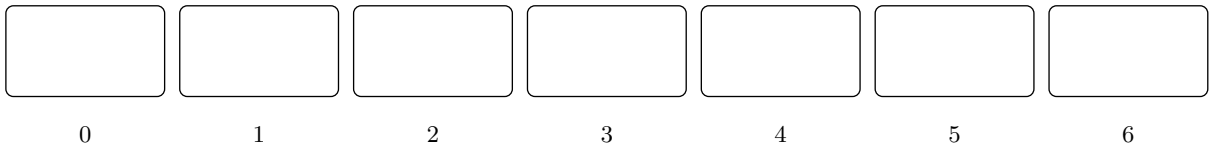




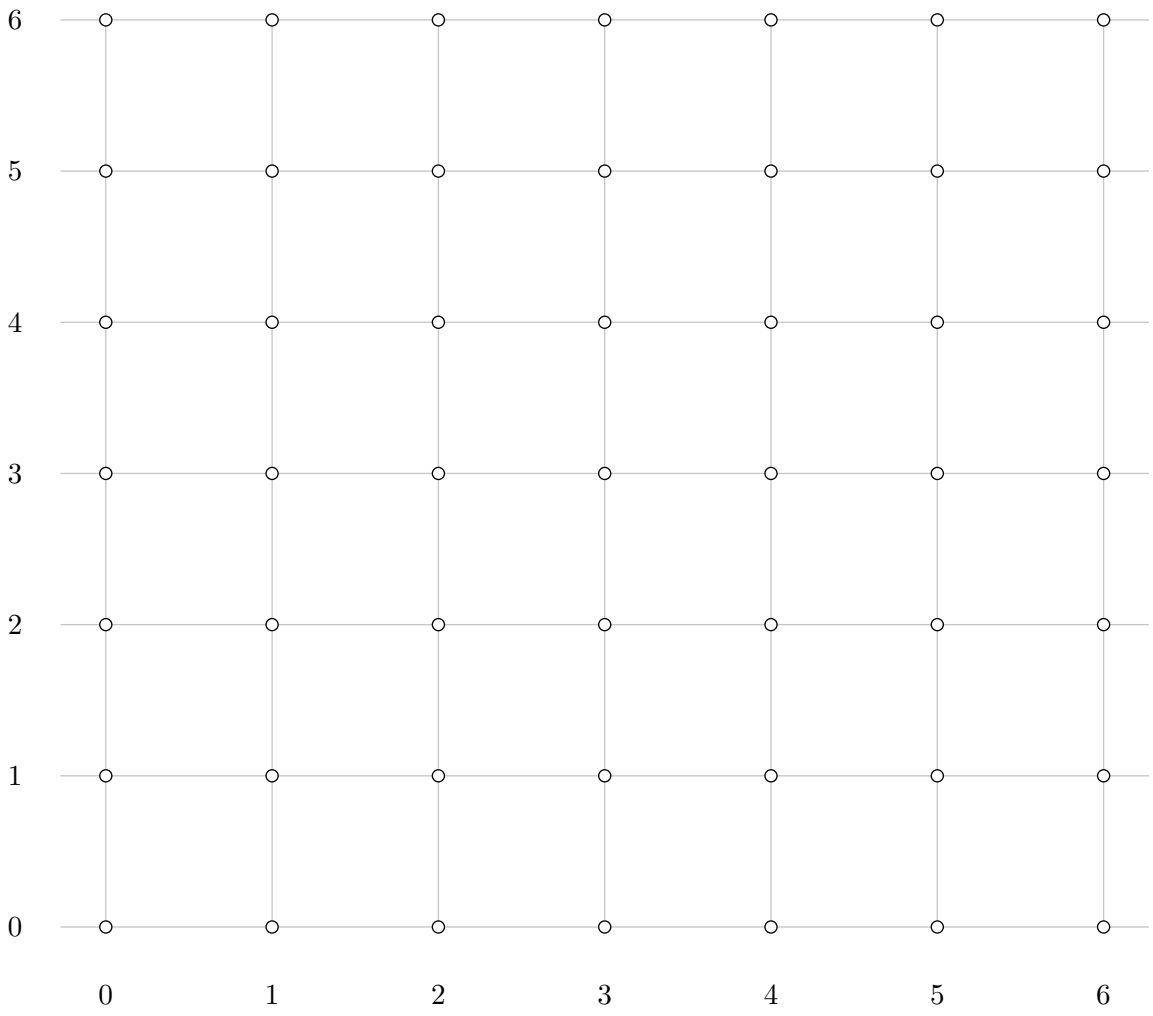
Problem 5: Make a landscape with the smallest total height, then one with the largest total height. Keep all three clues.







Problem 6: Choose two clues on the seven-site board that force exactly one whole landscape. Then choose two clues that allow more than one. Let a partner solve both.



Problem 7: Two clues are 4 steps apart. One is height 1. Which heights for the other clue are possible? Explain a rule for clues any number of steps apart.